

Uniqueness of the Rank-5 Projection in the Quantum Gauge Theory of Time

Pasquale Camelia*

April 2026

Abstract

We prove that $r = 5$ is the unique rank compatible with the foundational requirements of the QGT framework: Noetherian conservation, autosimilar closure, and operational metric stability. The proof is a two-sided exclusion with no free parameters.

Keywords. Non-invertible projection; rank uniqueness; Pincherle operator; boundary metric; autosimilar closure; Majorana spinor; Noether conservation; finite-rank operator; quantum gauge theory.

1 Introduction

The QGT programme derives physical structure from a single axiom: the existence of a conserved Noetherian flow under restricted accessibility. From this axiom, a sequence of forced steps leads to a non-invertible projection $\Pi : \mathcal{M}^5 \rightarrow \mathbb{R}^4$ of rank $r = 5$. The rank has been derived in three independent ways across the corpus.

Spectral curvature (P01). The Pincherle dilatation operator $\mathcal{D} = x d/dx$, acting on an irrational scale invariant φ , admits exactly five independent winding modes above the resolution threshold ($r = W = 5$).

Metric stability (P02). The boundary metric $d_{\partial}(f, g)^2 = \sum_{i=0}^4 \sigma_i^2 |c_i|^2$ is non-degenerate and operationally stable with exactly five retained singular modes.

Holonomy stability (QGT Foundations). The sector $n = 5$ is the unique structural equilibrium: non-Abelian closure is realised exactly once, without collapse or redundancy.

Each of these derivations establishes that $r = 5$ *works*. None of them, in isolation, proves that no other value of r could also work. The open problem declared in P06 is precisely this uniqueness statement:

The open problem declared in P06 is the following. Prove that the rank-5 projection is the *only* non-invertible projection from a Majorana bulk on $\mathcal{M}^5$ compatible with: (I) the conservation law $\partial_I J^I = 0$; (II) the autosimilar second-order closure; and (III) the requirement that the boundary be a stable, metrically consistent observable domain. Its proof would not change the physics; it would establish that the physics could not have been otherwise.

This paper closes that open problem: $r = 5$ is the *only* admissible finite rank of the QGT projection. All other ranks violate at least one of requirements (I)–(III).

*Independent Researcher, Italy. pasquale.camelia@gmail.com. ORCID: [0009-0006-4779-4647](https://orcid.org/0009-0006-4779-4647). © 2026 Pasquale Camelia. Licensed under [CC BY-NC 4.0](https://creativecommons.org/licenses/by-nc/4.0/) — free to share and cite with attribution; commercial use prohibited. Companion to: *The Quantum Gauge Theory of Time*, First Edition (KDP, ISBN 9798198195011). <https://pasqualecamelia.github.io>. Preprint DOI: [10.13140/RG.2.2.13475.62242](https://doi.org/10.13140/RG.2.2.13475.62242).

2 Three characterizations of $r = 5$

We recall the three results from the corpus that each independently point to $r = 5$. The notation follows the original papers.

2.1 Spectral curvature (P01)

The Fibonacci recursion $v_{k+2} = v_{k+1} + v_k$ has roots $\lambda_{\pm} = (1 \pm \sqrt{5})/2$ with squared separation $\kappa^2 = |\lambda_+ - \lambda_-|^2 = 5$. The autosimilar closure requires one resolvable singular direction per closure direction, forcing $r \geq W = 5$ [2].

2.2 Holonomy threshold (QGT Foundations)

Section 10.3 of [1] establishes that $n = 5$ is the unique holonomy sector at which the non-Abelian closure is realised exactly once without collapse or redundancy. For $n < 5$ the bilinear structure collapses; for $n > 5$ additional closure steps are redundant.

Proposition 2.1 (Holonomy threshold, [1] §10.3). *$n = 5$ is the minimal sector compatible with conservation, non-Abelian closure, and restricted accessibility. For $n < 5$: collapse. For $n > 5$: redundancy.*

2.3 Metric stability (P02)

The boundary metric $d_{\partial}(f, g)^2 = \sum_{i=0}^4 \sigma_i^2 |c_i|^2$ is non-degenerate and operationally stable with exactly five retained singular modes $\sigma_k = \varphi^{-k}$ [2]. The spectral resolution threshold α is derived independently from the Pincherle operator and Fibonacci curvature (independence remark in [2]); the bilateral inequality $w_4 > \alpha > w_5$, where $w_k = \sigma_k \sigma_{k+1} = \varphi^{-(2k+1)}$, is a verified consequence, not a definition.

3 Exclusion of $r \neq 5$

We now prove that every $r \neq 5$ violates at least one of requirements (I)–(III). We treat the cases $r < 5$ and $r > 5$ separately.

3.1 Ranks $r < 5$

Lemma 3.1 (Collapse for $r < 5$). *Let $r < 5$. Then any rank- r projection from a Majorana bulk with autosimilar structure φ either:*

- (a) *violates requirement (I) by failing to support a consistent Noetherian balance between accessible and coherent sectors; or*
- (b) *violates requirement (II) by being inconsistent with the second-order autosimilar closure.*

Proof. The Noetherian balance requires (QGT Foundations, §8.3):

$$\Delta Q_{\text{acc}} + \Delta Q_{\text{coh}} = 0. \tag{1}$$

For this balance to be non-trivially realised at the boundary, the projected sector must support at least as many independent modes as the spectral curvature demands.

From the second-order autosimilar closure (§2.1), the spectral curvature is $\kappa^2 = 5$, which demands $W = 5$ independent winding modes (Section 2). If $r < 5$, the projection retains fewer than W independent singular modes. Two cases arise:

Case (a): $r \leq 3$. Fewer than four independent modes are retained. The doublet structure required for the minimal non-Abelian invariance group $U(1) \times SU(2)$ cannot be realised: $SU(2)$ demands at least two degenerate modes plus one scalar mode, i.e. $r \geq 3$; and the full partition $(2+1+1+1)$ required for a stable non-Abelian sector demands $r \geq 4$. For $r \leq 3$, the invariance group collapses to $U(1)^r$ (Abelian), which cannot support the non-Abelian internal torsion required by requirement (II). Hence (II) is violated.

Case (b): $r = 4$. The partition $(2+1+1)$ supports $U(1) \times SU(2)$ but lacks the fifth mode required for the autosimilar closure to complete at $k = 4$. By Proposition 2.1, for $n < 5$ the bilinear structure collapses. The fifth closure step ($k = 4$, mode v_5) has no resolvable representative in the rank-4 truncation; the Noetherian balance (1) cannot be maintained. This violates (I).

In both sub-cases, either (I) or (II) fails. This proves the lemma. $\square$ $\square$

Remark 3.2 (Boundary metric for $r < 5$). For $r < 5$, even if one were to ignore the conservation and closure failures, the boundary metric would also degenerate via the bilateral threshold. For $r < 5$, the spectrum $\{\sigma_k = \varphi^{-k}\}_{k=0}^{r-1}$ does not close the Fibonacci recursion: the fifth mode $\sigma_4 = \varphi^{-4}$ is absent, so the recursion $v_{k+2} = v_{k+1} + v_k$ cannot be closed at $k = 4$. Equivalently, the transition weight $w_4 > \alpha$, where α is the independently derived threshold [2], meaning level $k = 4$ is operationally significant but excluded by the truncation. The boundary metric therefore fails to represent a resolvable transition that the projection structure requires, violating Requirement (III). This provides an independent confirmation of Lemma 3.1 without invoking any quantity that presupposes $r = 5$.

3.2 Ranks $r > 5$

Lemma 3.3 (Redundancy for $r > 5$). *Let $r > 5$. Then any rank- r projection from a Majorana bulk with autosimilar structure φ violates requirement (III): the boundary metric becomes redundant and operationally unstable.*

Proof. For $r > 5$, the projection retains more than $W = 5$ singular modes. We show this is incompatible with (III) in two independent ways.

(a) Spectral argument. The singular values of the autosimilar hierarchy decay as $\sigma_k = \varphi^{-k}$. The spectral resolution threshold α is derived in [2] (Remark 1) from the Pincherle operator and Fibonacci curvature, independently of any rank value. The transition weights $w_k = \sigma_k \sigma_{k+1} = \varphi^{-(2k+1)}$ satisfy $w_4 > \alpha > w_5$ as a verified consequence [2].

For $r > 5$, mode $k = 5$ has transition weight $w_5 < \alpha$: it lies below the resolution threshold and cannot sustain an independent observable partition. Including it assigns positive metric weight to an operationally invisible direction, violating Requirement (III).

(b) Holonomy argument. By Proposition 2.1(iii), for $n > 5$ additional closure steps generate no new independent bilinear invariants: the modes for $k \geq 5$ project onto the span of modes already present in the rank-5 sector. Retaining them in the boundary metric introduces redundant directions — configurations that are distinguished by the metric but produce identical boundary observables. This is precisely an operational instability: distinct r -dimensional configurations in $\mathcal{H}_{\text{obs}}$ map to the same physical observable, violating the non-degeneracy condition of (III).

Both arguments prove that $r > 5$ violates (III). $\square$ $\square$

Remark 3.4 (Connection to mode counting). The admissible mode set $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$ is determined by the condition $\pi(n) \leq r = 5$, where $\pi(n)$ is the prime-counting function. For $r > 5$, the condition $\pi(n) \leq r$ would admit $n \in \{2, 3, 5, 7, 11, 13\}$, but the mode $n = 13$ ($\pi(13) = 6$)

has no independent spectral contribution: the transition weight $w_5 < \alpha$ [2] confirms that no additional mode becomes resolvable for $r > 5$. The admissible set is therefore insensitive to $r > 5$ in terms of independent modes, confirming the redundancy argument.

4 The uniqueness theorem

We can now state and prove the main result.

Theorem 4.1 (Uniqueness of rank-5 projection). *Let $\Pi_r : \mathcal{M}^5 \rightarrow \mathbb{R}^4$ be a non-invertible, finite-rank projection from a Majorana bulk on $\mathcal{M}^5$ with autosimilar structure φ . Suppose Π_r satisfies simultaneously:*

- (I) *the Noetherian conservation law $\partial_I J^I = 0$ with its balance structure across accessible and coherent sectors;*
- (II) *the second-order autosimilar closure $v_{k+2} = v_{k+1} + v_k$, forced by non-Abelian residual transport under restricted accessibility;*
- (III) *the induced boundary metric d_{∂} is non-degenerate and operationally stable (all retained singular modes lie above the spectral resolution threshold).*

Then $r = 5$ is the unique admissible rank.

Proof. We show that the three requirements jointly force $r = 5$ by exclusion.

Step 1: $r \geq 5$ from (I) and (II). Requirement (II) forces the spectral curvature $\kappa^2 = 5$ (Section 2). The autosimilar closure requires one resolvable singular direction per closure direction, so $r \geq W = 5$ [2].

Independently, requirement (I) forces the Noetherian balance to be non-trivially realised across all five closure steps of the autosimilar hierarchy. Lemma 3.1 shows that for $r < 5$, either the balance fails (case a) or the closure structure cannot be realised (case b). Therefore $r \geq 5$.

Step 2: $r \leq 5$ from (III). For $r > 5$, Lemma 3.3 shows that the boundary metric includes singular mode $k = 5$ with transition weight $w_5 < \alpha$, below the independently derived threshold α [2]. This makes the metric assign positive distance to an operationally invisible direction, violating Requirement (III). Therefore $r \leq 5$.

Conclusion. Steps 1 and 2 give $r \geq 5$ and $r \leq 5$ simultaneously. Therefore $r = 5$. This value is admissible: it satisfies (I) by the Noether balance of Section 3, (II) by the autosimilar closure of [2] and [1], and (III) by the metric stability of Lemma 3.3. $\square$ $\square$

Corollary 4.2 (Forced structure of the observable boundary). *Under the hypotheses of Theorem 4.1:*

- *the effective rank is $r = 5$ with singular values $\sigma_k = \varphi^{-k}$, $k = 0, \dots, 4$;*
- *the boundary metric is $d_{\partial}^2 = \sum_{k=0}^4 \varphi^{-2k} |c_k|^2$;*
- *the structural threshold α satisfies $w_4 > \alpha > w_5$ [2];*
- *in the QGT angular readout convention, the rank-five boundary supports the admissible catalog $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$.*

No free parameter enters: all structures follow from requirements (I)–(III) together with the metric admissibility of [2].

Proof. The singular values $\sigma_k = \varphi^{-k}$ are forced by the autosimilar closure $v_{k+2} = v_{k+1} + v_k$ and normalization $\sigma_0 = 1$. The threshold α is derived independently of the rank in [2]. In the QGT angular readout convention, the rank-five boundary is associated with prime angular denominators satisfying $\pi(n) \leq 5$, giving $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$. $\square$ $\square$

5 Status and scope

5.1 What is established

Theorem 4.1 closes the open problem declared in P06. The proof is a corollary of three results established independently in the corpus: the spectral curvature mechanism [2], the holonomy threshold [1], and the metric stability analysis [2], with the spectral threshold derived from first principles in [2] and applied here without circularity. Every downstream result whose only open dependence was the uniqueness of $r = 5$ is therefore closed, conditionally within the Majorana–Fibonacci projective class.

5.2 Scope

The theorem takes as inputs the Majorana condition and the autosimilar structure φ ; it does not re-derive them. The uniqueness of $\mathcal{M}^5 = \mathbb{R}^4 \times S^1$ is established separately in QGT Foundations. The three open problems that remain after this closure — the CMB transfer function $G_{\text{edge}}^{\eta\xi}(k, \ell)$, the Yukawa map $\Phi_Y : H_{\text{ker}} \rightarrow \{m_e, m_\mu, m_\tau\}$, and the near-identity $e\pi$ correction — are unaffected.

6 Discussion

The uniqueness of $r = 5$ has a direct physical reading. The gauge group $G = U(1) \times SU(2) \times SU(3)$, the admissible mode set $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$, the fine-structure constant α , and the electroweak mixing angle $\sin^2 \theta_W = 3/13$ were all derived under the assumption $r = 5$. The theorem proves that, within the Majorana–Fibonacci projective class and the metric admissibility conditions of [2], this was not an independent assumption: it was the only consistent possibility.

The proof is a two-sided exclusion: requirements (I) and (II) impose $r \geq 5$; requirement (III) imposes $r \leq 5$. The three requirements are independent — (I) is dynamical, (II) algebraic, (III) geometric — and their convergence on the same bound is the structural content of the theorem.

7 Conclusions

We have proved that the rank-5 projection is the *only* non-invertible projection from a Majorana bulk on $\mathcal{M}^5$ compatible with the three foundational requirements of the QGT framework: Noetherian conservation, autosimilar second-order closure, and operational metric stability.

Spectral curvature forces $r \geq 5$; holonomy collapse excludes $r < 5$; metric redundancy excludes $r > 5$. Their conjunction yields $r = 5$ as the unique solution. The rank, and all structures derived from it, are consequences of the conservation axiom under restricted accessibility, within the Majorana–Fibonacci projective class and the metric admissibility conditions of [2].

Declarations. All results are analytic. The author declares no competing interests and received no funding.

References

- [1] P. Camelia, *A Quantum Gauge Theory of Time (QGT): Foundations from a Noetherian Balance*, Preprint (2026). https://pasqualecamelia.github.io/papers/Camelia_QGT_Foundations_of_Physics.pdf

- [2] P. Camelia, *Fibonacci-Autosimilar Compact Projection Maps: A Rank-Five SVD Characterization*, Preprint (2026). https://pasqualecamelia.github.io/papers/Camelia_2026_ProjectionSVD_BoundaryMetric.pdf